

# DRIFT

## A trading bot whose risk rules you can verify

Off-chain quant research · on-chain risk gate on BNB Chain

Live demo

[drift-macroguard.vercel.app/macroguard](https://drift-macroguard.vercel.app/macroguard)

Dex's contract on BSC Testnet

[0x8b09ebB85Be8Ed55Bb5132d29eABc567c42aa83D](https://bscscan.com/address/0x8b09ebB85Be8Ed55Bb5132d29eABc567c42aa83D)

Indonesia Web3 Hackathon 2026 · Dex Bennett

# The problem

Trading bots ask users to trust risk rules nobody can see.

Settings change silently; off-chain decision logs can be rewritten.

# The promise

Anyone can check what DRIFT's bot is allowed to do right now,  
and every recorded decision is public on BscScan.

# Who needs it, and how it could pay

**HYPOTHESIS:** not yet validated with users; no market figures claimed

## Who needs it

Retail traders who follow or rent bots: proof that the risk stop is real

Bot builders and signal sellers: a record of discipline a screenshot cannot fake

Communities and auditors: check a bot's rules without trusting its operator

## How it could pay

A risk gate per bot: its own MacroGuard plus a public panel, as a monthly service

The panel link as a trust badge for strategy sellers

No custody and no performance fee; on-chain cost per decision stays tiny (slide 7)

# How DRIFT works

## 1 Signal

Python engine: four strategies, point-in-time, no look-ahead

## 2 Ask the gate

MacroGuard.allowed(signal) on BSC Testnet: regime + 20% halt

## 3 Trade

Runner places the order on Bybit testnet (vetoed signals go Flat)

## 4 Record

recordDecision(...) leaves a public receipt; a breach halts the contract

Periwinkle: off-chain engine · white: on-chain contract

The contract never executes orders; it is the public rule and the public record.

# The panel, and asking the contract

DRIFT

How it works

MacroGuard

Contact

Open cockpit

MACROGUARD · BNB SMART CHAIN TESTNET · CHAIN 97

What is the bot allowed to do right now?

DRIFT's quant engine runs off-chain. Its risk gate lives in a public contract. This page shows the contract's state as read from BNB Chain. It never signs or sends a transaction.

read at 13:51:24 via data-seed-prebsc-1-s1.bnbchain.org:8545

Refresh

LIVE RISK STATE

Running — no drawdown halt

Open. The contract allows Long, Short and Flat right now.

Market regime stored on-chain: Neutral

Risk off  
Long vetoed

Neutral  
All signals open

Risk on  
All signals open

Who changes the regime?  
Only the agent key, via setRegime, from the engine's macro classifier.

What triggers a halt?  
A recorded decision with drawdown past the threshold. The contract flips it itself.

ON-CHAIN PROOF

LIVE READ · PUBLIC RPC · BLOCK 134,393,170

Contract  
0x8b09eb88...c42aa830

Agent (only key that can write)  
0x2B07AFB5...C6714b81

Halt threshold  
20% (2000 bps)

Decisions logged  
2

Open on BscScan

Deploy tx

WHY A SIGNAL IS ALLOWED OR BLOCKED

contract call allowed (signal)

LONG  
Allowed  
Open while the regime is not Risk off.

SHORT  
Allowed  
Only a drawdown halt would block it.

FLAT  
Allowed  
Always allowed — it reduces exposure.

DECISION TRAIL · VERIFIED RECEIPTS

smoke test 2025-09-30 - every row opens BscScan

01 Deploy  
constructor(maxDrawdownBps = 2000)

Neutral · not halted · 0 decisions

status 1 #133,995,398 · 0x2d8cce...c044

Live regime, halt, 20% line and decisions,  
read from BSC Testnet at a named block

ASK THE CONTRACT · SIMULATION

eth\_call · nothing signed, nothing written

What would MacroGuard decide if the bot reported this?

Pick a signal and the drawdown the bot would report. Your browser sends one eth\_call of recordDecision to the live contract, as if from the agent address. It is a simulation: nothing is signed or written, no transaction is created and no gas is spent. The answer comes from the contract on BSC Testnet, not from this page.

Signal the bot wants

Long  
buy

Short  
sell short

Flat  
exit

Drawdown it reports

-25.0% (-2,500 bps)

0% -20% halt line -30%

Ask the live contract

Blocked Long at -25.0%

At -25.0% the drawdown is at or past the 20% halt line, so the contract halts itself first. After a halt only Flat passes.

```
call recordDecision("BTCUSD*", Long, 0, -2500)
from agent 0x2B07_4b81 · unsigned
to MacroGuard 0x8b09_a830
read at block 134,602,182 via data-seed-prebsc-1-s1.bnbchain.org:8545
```

Simulation only: the node ran the call and discarded the result. The on-chain decision count is unchanged.

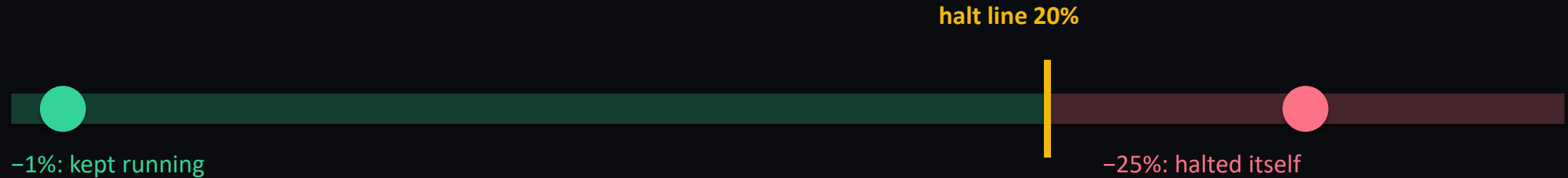
▼ Replay it with Foundry's cast

```
cast call 0x8b09eb8859e0Ed558b5132d29eABc567c42aa830
'recordDecision(string,uint8,uint256,int256)(bool)*' --from
0x2B07AFB54068042664074781AF36163aC6714b81 --rpc-url https://bsc-
testnet-rpc.publicnode.com -- BTCUSD 1 0 -2500
```

copy

Long at -25%: Blocked, answered by the live contract  
via eth\_call from the agent address (simulation)

# -25% recorded: the contract halted itself



|              |                                  |                                          |
|--------------|----------------------------------|------------------------------------------|
| #134,042,196 | setRegime(RiskOff)               | Long blocked; Short and Flat open        |
| #134,042,256 | recordDecision(BNB, Short, -1%)  | Decision #1, inside the line             |
| #134,042,283 | recordDecision(BNB, Short, -25%) | Halted event: only Flat allowed          |
| #134,042,318 | resume()                         | Agent-only, on the record                |
| #134,042,328 | setRegime(Neutral)               | Long allowed again; 2 decisions on-chain |

Every row is a transaction with receipt status 1 on BscScan · block 134,042,283 is the halt

# Why BNB Chain

**$\approx 0.0000034$  tBNB**

per recorded decision: 33,189–34,323 gas  
at 0.1 gwei (our receipts, BSC Testnet)

**$\approx 0.00008$  tBNB a day**

for an hourly bot (24 decisions);  
deploy: 449,207 gas  $\approx 0.000045$  tBNB

## What else it gives us

EVM: Foundry tests and Sourcify verification on chain 97

Public RPCs a browser can call: the panel needs no backend

Five smoke-test transactions within 132 blocks

Mainnet (chain 56) is a config switch; not deployed there

Testnet gas price, not a mainnet quote. Mainnet is on the roadmap, after an audit.

# Evidence and current limits

30 / 30

contract tests: unit, fuzz, invariant  
100% coverage of MacroGuard.sol

47

offline engine tests;  
no look-ahead tested as a property

15

web tests: contract reads and the  
what-if encoder against cast

exact match

source verified on Sourcify  
6 receipts, all status 1

## Limits, said plainly

The contract never executes orders · the runner fails open if the RPC is down · the agent can resume() at once  
The drawdown is reported by the agent · no live bot tick and no profit claimed; backtests are research

# What Dex added

## Upstream DRIFT project

Quant engine, cockpit and MacroGuard.sol

7 contract tests

Built for a Mantle hackathon track (June 2026)

Migrated to BNB Chain in bcc-ukdw/seed-bnb  
(commit 52671ce, 29 Sep 2026)

## Dex's work in this fork

Own BSC Testnet deploy + 5-tx smoke test (status 1)

Source verified on Sourcify (exact match)

Public panel: live browser reads, receipts, limits

Ask the contract: eth\_call what-if, nothing signed

+23 contract tests, 47 engine tests, 15 web tests, CI

Labelled Binance data fallback · threat model

Landing, visual system, PRD, this deck

Code: [github.com/Stylenecy/seed-bnb/tree/dex/drift/drift](https://github.com/Stylenecy/seed-bnb/tree/dex/drift/drift)

# Roadmap

Not built yet; listed in order of trust gained per step

Resume through a multisig or timelock

Fail-closed mode for the runner

Signed regime verdicts (EIP-712) with an input hash

Attested equity, so the drawdown is not self-reported

Record the raw signal plus a veto flag

Agent identity (ERC-8004), then mainnet after an audit

## Check it yourself

[drift-macroguard.vercel.app/macroguard](https://drift-macroguard.vercel.app/macroguard)

[github.com/Stylenecy/seed-bnb/tree/dex/drift/drift](https://github.com/Stylenecy/seed-bnb/tree/dex/drift/drift)

BscScan: [0x8b09ebB8...2aa83D](https://bscscan.com/address/0x8b09ebB8...2aa83D)

Sourcify: [repo.sourcify.dev/97/...](https://repo.sourcify.dev/97/...)